Band-1 Lehmer / Mahler Measure Probe (Collapse List 1.11)

Christopher Baird + ZoraASI Band-1 dash (executor)

2026-09-11

Band-1: Lehmer Pairs / Mahler-Measure Next Examples (Probe Only)

Honesty stamp. This note does **not** claim that Lehmer's conjecture is solved, that τ is proved minimal, that any Clay / Millennium stamp applies, or that a Theory of Everything is finished. **GATE-04 collection remains LOCKED** (out of scope). A 2026 Zenodo deposit claiming Mahler measures below Lehmer's τ is ledger-noted and **not** elevated above the peer-reviewed / Mossinghoff-aligned champion.

Lehmer's conjecture remains OPEN. Box certificates $\neq$ solution.

Collapse-list target: **1.11 — Lehmer pairs / Mahler-measure next examples**.

Abstract

1. **Literature champion:** Lehmer's number

$$\tau = M(L) \approx 1.176280818259917\ldots$$

for

$$L(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1,$$

remains the smallest known Mahler measure > 1 in peer-reviewed surveys and Mossinghoff's database (cutoff **2026-09-11**).
2. **Recent work (2024–2026):** El-Serafy–McKee (*Canad. Math. Bull.* **68**, 2025) completeness under house/shortness bounds (236 known "tiny" measures in $(1, 1.25]$); McKee–Smyth (arXiv:2605.19013) classify all length-5 Salem polynomials and structure length-6; Sac-Épée adds Salem numbers near θ_0.
3. **Box:** Mahler measures for 14 integer polynomials (cyclotomic exclusions + Lehmer sanity + nearby Salem/Pisot checks). All cyclotomics give $M = 1$; box recovers $M(L) \approx 1.17628081825992$.

No new record below τ is claimed.

Definitions

For $P(x) = a_n \prod_i (x - \alpha_i) \in \mathbb{Z}[x]$,

$$M(P) = |a_n| \prod_i \max\bigl(1, |\alpha_i|\bigr).$$

Kronecker: $M(P) = 1$ iff P is a product of cyclotomics (up to units). **Lehmer's question:** does there exist $C > 1$ such that $M(P) > 1 \Rightarrow M(P) \geq C$?

Literature ledger (survey 2026-09-11; large searches cited, not re-run)

Year	Authors	Item	Status
1933	Lehmer	$L(x)$; $\tau \approx 1.17628\ldots$; gap question	classical champion
1977	Boyd	Small Salem tables	published
1971/79	Smyth	Non-reciprocal: $M \geq \theta_0 \approx 1.3247\ldots$	theorem
1998–	Mossinghoff	Tables / DB: deg ≤ 200, $M < 1.3$	published + database
2024	Sac-Épée	+11 Salem in $(1.3, 49/37)$ outside τ_n family	arXiv / in press
2025	El-Serafy–McKee	House/shortness completeness; 236 tiny	*CMB* 68
2026	McKee–Smyth	Length-5 complete; length-6 = 12 families + 126 sporadics (≥ 1.01)	arXiv:2605.19013
2026	Helmdach (Zenodo)	Claim $M_{31} \approx 1.03878 < \tau$	**non-canonical — not elevated**

Canonical champion. Peer-reviewed and database consensus still list $\tau = M(L)$ as the smallest known $M > 1$. Informal Zenodo claims below τ receive the same hygiene as Goldbach Zenodo claims past the Math. Comp. verification bound.

Method (box)

1. Roots via `numpy.roots`; product of $\max(1, |\alpha_i|)$ in `mpmath` (dps=50).
2. List: $\Phi_1, \ldots, \Phi_6$ (expect $M = 1$); $x^2 - x - 1$; $x^3 - x - 1$ (θ_0); Lehmer minimal L; Lehmer length-5 short form $z^{12} - z^7 - z^6 - z^5 + 1$; two nearby Salem shorts (≈ 1.200, ≈ 1.216); $x^2 - 2$; $x - 2$.
3. Label: sanity / exclusion probe — **not** an exhaustive record hunt.

Results (box-certified tonight)

Polynomial	Class	Box M	OK
$\Phi_1, \ldots, \Phi_6$	cyclotomic	**1**	yes
$x^2 - x - 1$	Pisot	≈ 1.61803398875	yes
$x^3 - x - 1$ (θ_0)	Pisot	≈ 1.32471795724	yes
Lehmer minimal L	Salem record	≈ 1.17628081826	sanity
Lehmer short len-5	Salem$\times\Phi_6$	≈ 1.17628081826	yes
Salem ≈ 1.200 short	Salem	≈ 1.20002652399	yes
$z^{10} - z^6 - z^5 - z^4 + 1$	Salem	≈ 1.21639166114	yes
$x^2 - 2$, $x - 2$	other/Pisot	2	yes

- Cyclotomic exclusions all $M = 1$: **True**
- Lehmer sanity vs 1.1762808183: **True**
- Wall ≈ 0.004 s; $n = 14$ polynomials
- Artifacts: `experiments/band1/lehmer_mahler/results.json`

What is still OPEN (cited next)

1. **Global minimality of τ:** is $M(L)$ the infimum of $\{M(P) : P \in \mathbb{Z}[x],\ M(P) > 1\}$? (Lehmer conjecture; strong form for Salem numbers.)
2. **Any example with $1 < M < \tau$:** none known in peer-reviewed literature.
3. **Complete tiny-measure census** beyond El-Serafy–McKee shortness/house hypotheses; extend Mossinghoff tables.
4. McKee–Smyth length-6 classification below house 1.01 without assuming Lehmer (Dobrowolski gap makes full push impractical).

Non-claims

- Lehmer conjecture **not** solved.
- Box did **not** find / certify any M in $(1, \tau)$.
- Zenodo 2026 Helmdach family **not** treated as the new record.
- No Clay stamp; no finished ToE; **GATE-04 LOCKED**.

References (short)

- D.H. Lehmer, *Ann. of Math.* (2) **34** (1933), 461–479.
- M.J. Mossinghoff, *Math. Comp.* **67** (1998); Mahler-Measure-Database.
- S. El-Serafy, J. McKee, *Canad. Math. Bull.* **68** (2025), 603–619.
- J. McKee, C. Smyth, *Short Salem polynomials*, arXiv:2605.19013 (2026).
- J.-M. Sac-Épée, arXiv:2409.11159; Rocky Mountain J. Math. (in press).
- C.J. Smyth, survey in *Number Theory and Polynomials* (CUP, 2008).

Paths

- Lab: `/workspace/mission-control/experiments/band1/lehmer_mahler/`
- Paper: `papers/2026-09-11_Band1_Lehmer_Mahler_measure_probe.md` (+ PDF)
- Results: `results.json` · Ledger: `literature_ledger.json`